

✓ Student Attitudes Toward Mobile Phone Regulation in Schools

Analysis using PISA 2022 data (Argentina)

SECTION 1. RESEARCH QUESTION

What do students think about banning mobile phones in school?

SECTION 2. DATASET

Source: PISA 2022 student questionnaire

Country: Argentina

Variables: IC179Q01 – IC179Q06

Topic: regulation of digital devices in school

SECTION 3. DATA PREPARATION

```
from google.colab import drive
drive.mount('/content/drive')
```

Mounted at /content/drive

```
!unzip -l "/content/drive/MyDrive/PISA_2022_1.zip"
```

```
Archive: /content/drive/MyDrive/PISA_2022_1.zip
  Length      Date    Time    Name
  -----
2096666903  2024-05-28  09:52    CY08MSP_STU_QQQ.SAV
  -----
2096666903                      1 file
```

```
!pip install pyreadstat
```

Collecting pyreadstat

Downloading pyreadstat-1.3.3-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (3.1 MB)
Requirement already satisfied: narwhals>=2.10.1 in /usr/local/lib/python3.12/dist-packages (from pyreadstat) (2.10.1)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (from pyreadstat) (2.0.2)
Downloading pyreadstat-1.3.3-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (3.1/3.1 MB) 81.2 MB/s eta 0:00:00

Installing collected packages: pyreadstat
Successfully installed pyreadstat-1.3.3

```
!pip install pyreadstat
```

Requirement already satisfied: pyreadstat in /usr/local/lib/python3.12/dist-packages (1.3.3)
Requirement already satisfied: narwhals>=2.10.1 in /usr/local/lib/python3.12/dist-packages (from pyreadstat) (2.10.1)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (from pyreadstat) (2.0.2)

```
import pyreadstat
```

Variables used

IC179Q01JA Students should not be allowed to bring mobile phones to class

IC179Q03JA Students should collaborate with teachers to define rules for digital devices

IC179Q04JA School should filter access to social media

IC179Q05JA School should block online games

IC179Q06JA Teachers should monitor what students do on their laptops

Response scale

- Strongly disagree
- Disagree
- Agree
- Strongly agree

```
import pandas as pd

cols = [
    "CNT",
    "OECD",
    "IC179Q01JA",
    "IC179Q02JA",
    "IC179Q03JA",
    "IC179Q04JA",
    "IC179Q05JA",
    "IC179Q06JA"
]

df = pd.read_spss("/content/pisa/CY08MSP_STU_QQQ.SAV", usecols=cols)

arg = df[df["CNT"] == "Argentina"]

len(arg)

12111

df = df.rename(columns={
    "IC179Q01JA": "ban_mobile_phones",
    "IC179Q03JA": "rules_with_teachers",
    "IC179Q04JA": "filter_social_media",
    "IC179Q05JA": "block_games",
    "IC179Q06JA": "teacher_monitoring"
})

arg["ban_mobile_phones"].value_counts(normalize=True)*100
```

	proportion
IC179Q01JA	
Strongly disagree	41.406359
Disagree	39.333240
Agree	14.007564
Strongly agree	5.252837

dtype: float64

```
arg["rules_with_teachers"].value_counts(normalize=True)*100
```

	proportion
IC179Q03JA	
Agree	55.641688
Disagree	18.145277
Strongly agree	15.144990
Strongly disagree	11.068045

dtype: float64

```
arg["filter_social_media"].value_counts(normalize=True)*100
```

	proportion
IC179Q04JA	
Disagree	40.020501
Agree	29.374725
Strongly disagree	22.624103
Strongly agree	7.980671

dtype: float64

```
arg["block_games"].value_counts(normalize=True)*100
```

	proportion
IC179Q05JA	
Agree	35.891416
Disagree	34.262656
Strongly disagree	21.041820
Strongly agree	8.804109

dtype: float64

```
arg["teacher_monitoring"].value_counts(normalize=True)*100
```

	proportion
IC179Q06JA	
Agree	41.749416
Disagree	26.664720
Strongly disagree	18.647780
Strongly agree	12.938084

dtype: float64

```
support = {  
    "Prohibir celulares": (  
        (arg["IC179Q01JA"] == "Agree").mean() +  
        (arg["IC179Q01JA"] == "Strongly agree").mean()  
    ) * 100,
```

```

"Reglas con docentes": (
    (arg["IC179Q03JA"] == "Agree").mean() +
    (arg["IC179Q03JA"] == "Strongly agree").mean()
) * 100,

"Monitoreo docente": (
    (arg["IC179Q06JA"] == "Agree").mean() +
    (arg["IC179Q06JA"] == "Strongly agree").mean()
) * 100
}

```

```

import pandas as pd
import matplotlib.pyplot as plt

support = pd.Series({
    "Prohibir celulares": 19.260401,
    "Reglas con docentes": 70.786678,
    "Monitoreo docente": 54.687500,
    "Prohibir juegos online": 44.695525
})

fig, ax = plt.subplots(figsize=(9, 5.2))

bars = ax.barh(support.index, support.values)

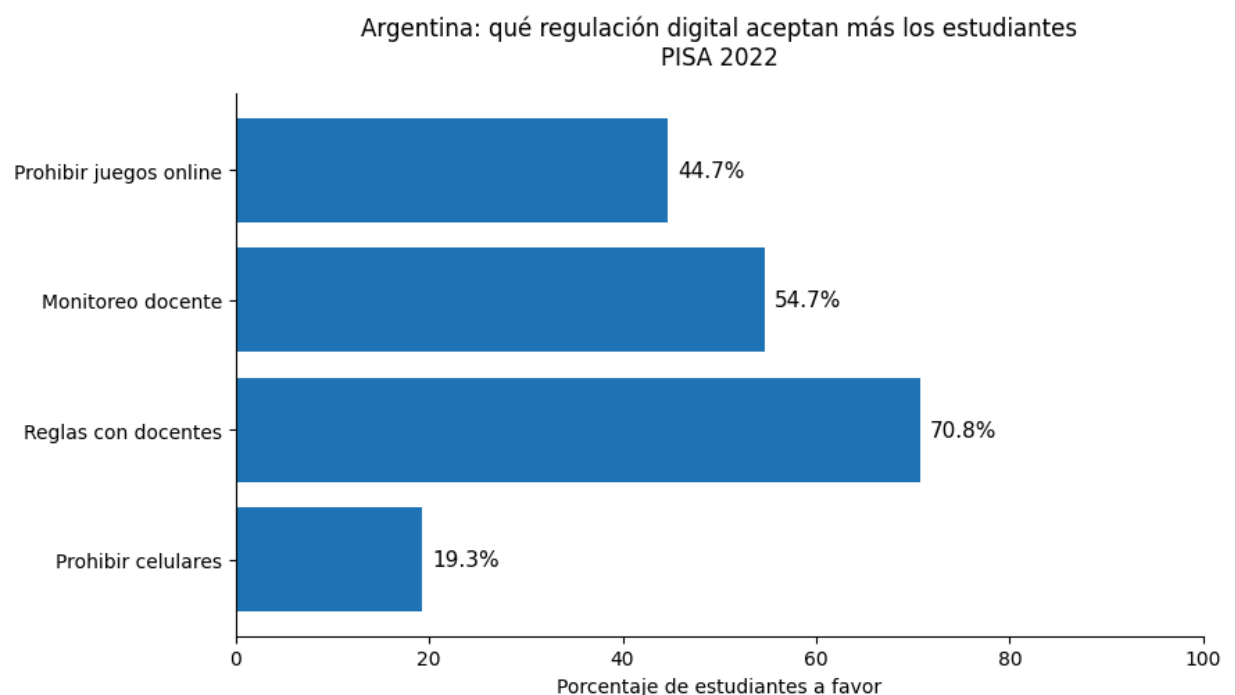
ax.set_xlim(0, 100)
ax.set_xlabel("Porcentaje de estudiantes a favor")
ax.set_title("Argentina: qué regulación digital aceptan más los estudiantes\nPISA 2022", pad=15)

for i, v in enumerate(support.values):
    ax.text(v + 1, i, f"{v:.1f}%", va="center", fontsize=11)

ax.spines["top"].set_visible(False)
ax.spines["right"].set_visible(False)

plt.tight_layout()
plt.show()

```



SECTION 4. RESULTS

81% of students oppose banning mobile phones in school.

However, 71% support creating rules together with teachers.

SECTION 5. INTERPRETATION

Students appear to reject total bans, but they show support for rules and teacher supervision regarding device use.

POTENTIAL EXTENSION

Relationship between student attitudes toward phone bans and academic performance.